SHIH-WEN LIU (CASPER)

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Research Interests

My core interests are **cooperative and scalable robotic intelligence**. I particularly focus on designing learning mechanisms that enable multiple robots to coordinate and collaborate efficiently, especially in multi-agent robotic manipulation and control. Previously, I worked on parameter-efficient fine-tuning for multi-task learning, and histopathology report generation.

Education

- **National Cheng Kung University** — *Sep. 2024 – Present*
  Master of Science in Artificial Intelligence (Advisor: Prof. Wei-Ta Chu) — Tainan, Taiwan
  - CGPA: 4.00/4.00
- **National Sun Yat-sen University** — *Sep. 2020 – Jun. 2024*
  Bachelor of Science in Computer Science Engineering — Kaohsiung, Taiwan
  - CGPA: 3.62/4.00, Academic Excellence Award (2020)

Publications

[C.2] **Shih-Wen Liu**, Yen-Chang Chen, Wei-Ta Chu, Fu-En Yang, Yu-Chiang Frank Wang "**Frequency Switching Mechanism for Parameter-Efficient Multi-Task Learning**." In *Proceedings of the IEEE/CVF International Conference on Computer Vision (**CVPR**) 2026. (Work done in collaboration with NVIDIA.)*

[C.1] **Shih-Wen Liu**, Hsuan-Yu Fan, Wei-Ta Chu, Fu-En Yang, Yu-Chiang Frank Wang "**Histopathology Image Report Generation by Vision Language Models with Multimodal In-Context Learning**." In *the Conference of Medical Imaging with Deep Learning (**MIDL**) 2025. (Work done in collaboration with NVIDIA.)*

Projects

- **SUIII! A Ball Focus and Identity Memory for Robust Soccer Player Tracking** — *Feb. 2025*
  Tools: Python, PyTorch — [Project]
  - Improved soccer player tracking under occlusion using ball-focused reasoning, group motion dynamics, and long-term appearance memory
- **TRUMP: Task-Responsive Unified Multi-modal Perception** — *Sep. 2024*
  Tools: Python, PyTorch — [Project]
  - Developed a task-adaptive perception framework that extracts task-aware visual features from natural-language task descriptions
  - Designed a hypernetwork to generate additional task vectors for flexible task-specific adaptation
- **Split Conv: Efficient Convolution Layer Design** — *Aug. 2024*
  Tools: Pytorch — [Project]
  - Designed memory-efficient convolution and pooling layers for large-resolution training
  - Optimized implementations to reduce VRAM usage by 50 times
- **Jetson Nano Real-Time AR Teaching System** — *Jun. 2024*
  Tools: Python, OpenCV, Jetson — [Demo Video]
  - Developed AR overlay pipeline on Jetson Nano for live video processing
  - Achieved 30 FPS on 720p streams with real-time object detection
- **Desiary - A Novel Diary-Dating App** — *May 2024*
  Tools: Flask, React Native, Stable Diffusion, LoRA, MongoDB, Docker — [Demo Video]
  - Created diary-to-image generation using Stable Diffusion with LoRA fine-tuning
  - Built full-stack mobile app with user authentication and image gallery
  - Deployed backend in Docker with CI/CD pipelines for reliability
- **Newsbie - Your Little News Reporter** — *May 2024*
  Tools: Python, Flask, React Native, MongoDB, Docker — [Demo Video]
  - Implemented web-scraping modules to aggregate news from 10+ sources
  - Integrated LLM API for concise, on-demand article summarization and podcast generation pipeline
- **Let the Particles Have Babies! - PSO + GA + NN Mapping** — *Feb. 2024 - Jun. 2024*
  Tools: Python — [Project]
  - Combined Particle Swarm Optimization and Genetic Algorithms with neural networks for search-space reconstruction
- **Real-Time Multi-Filter App** — *Nov. 2022 - Dec. 2023*
  Tools: Python, NumPy — [Project]

◦ Developed 20+ stackable image filters (emboss, oil paint, pencil sketch) in pure NumPy
◦ Optimized vectorized operations to maintain 60+ FPS on 1080p frames

- **Glomerular Detection & Disease Classification** *Nov. 2022 - Dec. 2023*
  Tools: Python, PyTorch [Report Video]
  ◦ Accelerated segmentation pipeline from 364 to 29 days via entropy-based filtering
- **Taiwanese Language Learning App Based on LLMs** *Sep. 2022 - Mar. 2024*
  Tools: Unity, Python, SpeechRecognition, TTS [Project]
  ◦ Integrated Unity-based AR with an LLM-driven agent for immersive language-learning scenarios
  ◦ Leveraged vision-language action commands to control interactive virtual agent
  ◦ Developed real-time speech recognition and synthesis pipelines for pronunciation feedback

Experience

- **TAICA-Introduction to Artificial Intelligence [🌐]** *Fall 2025*
  Teaching Assistant Tainan, Taiwan
  ◦ Selected course under the TAICA Program with students from 50+ universities.
  ◦ Assisted in courses and assignments design.
- **NCKU CSIE-Linear Algebra [🌐]** *Spring 2024*
  Teaching Assistant Tainan, Taiwan
  ◦ Assisted in courses and graded assignments.
- **NXP Semiconductors [🌐]** *Feb. 2023 – Aug. 2023*
  Industry Intern Kaohsiung, Taiwan
  ◦ Developed automated data preprocessing tools, algorithms, and logic validation scripts for large-scale data.
  ◦ Enhanced existing pipelines to improve performance and maintainability.
- **Scratch Programming Instructor** *Summer 2021 & 2022*
  Instructor Hsinchu, Taiwan
  ◦ Taught introductory Scratch programming to elementary students, boosting communication and teaching skills.

Honors and Awards

- **Best Team Award** *2024*
  Critical Care Data Science Conference & Taiwan Datathon
  ◦ Recognized for outstanding team collaboration and high-performance ICU data analysis with senior clinicians.

Skills

- **Programming Languages:** Python, C/C++, JavaScript, HTML, CSS
- **Data Science & Machine Learning:** NumPy, Pandas, PyTorch, TensorFlow, scikit-learn, Matplotlib
- **Developer Tools:** Docker, ROS2, Unity, React Native, Git, Bash
- **Specialized Areas:** Parameter-Efficient Fine-Tuning (LoRA), Adaptive Multi-Task Vision Models, Large Multi-Modal Model, App development